



**PowerPoint Presentation by**  
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# **MANAGEMENT ACCOUNTING**

**8<sup>th</sup> EDITION**

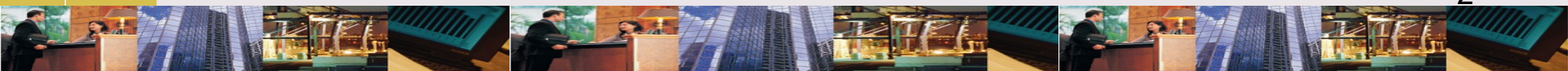
**BY**

**HANSEN & MOWEN**

## **10 SEGMENTED REPORTING**

# ***LEARNING OBJECTIVES***

After studying this chapter, you should be able to:



# ***LEARNING OBJECTIVES***

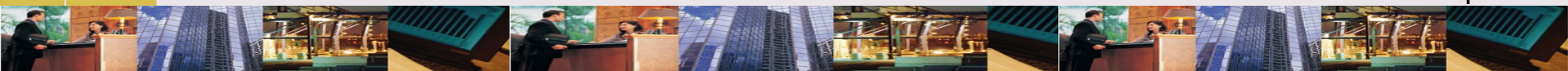
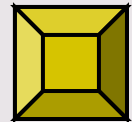
1. Explain how & why firms choose to decentralize.
2. Explain the difference between absorption & variable costing, & prepare segmented income statements.
3. Compute & explain return on investment (ROI).

Continued

# ***LEARNING OBJECTIVES***

4. Compute & explain residual income & economic value added (EVA).
5. Explain the role of transfer pricing in a decentralized firm.

*Click the button to skip  
Questions to Think About*



# QUESTIONS TO THINK ABOUT: Galactic-Media Inc.

Why do firms calculate  
income? What information  
does it provide?

# QUESTIONS TO THINK ABOUT: Galactic-Media Inc.

What costs go into  
inventory? How can they  
affect income?

# QUESTIONS TO THINK ABOUT: Galactic-Media Inc.

What is GAAP, & how does it affect the income statement of the Medical Supplies Division?

# QUESTIONS TO THINK ABOUT: Galactic-Media Inc.

What do you suppose Kathy's chances are for getting the vice president to consider evaluating her performance on the basis of variable, instead of absorption, costing?



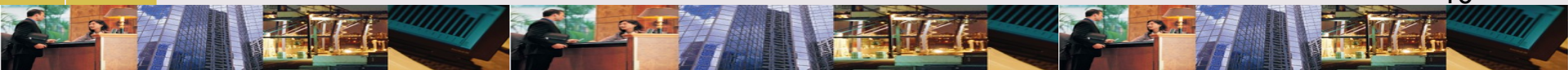
# ***LEARNING OBJECTIVE***

***1***

Explain how & why  
firms choose to  
decentralize.

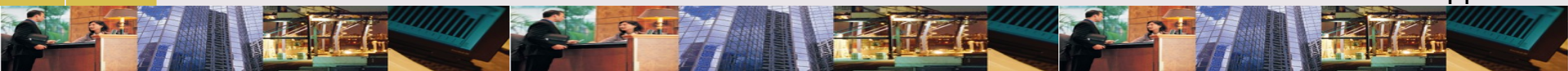
What is a responsibility accounting system?

**A responsibility accounting system** measures the results of responsibility centers according to information managers need to operate their centers.

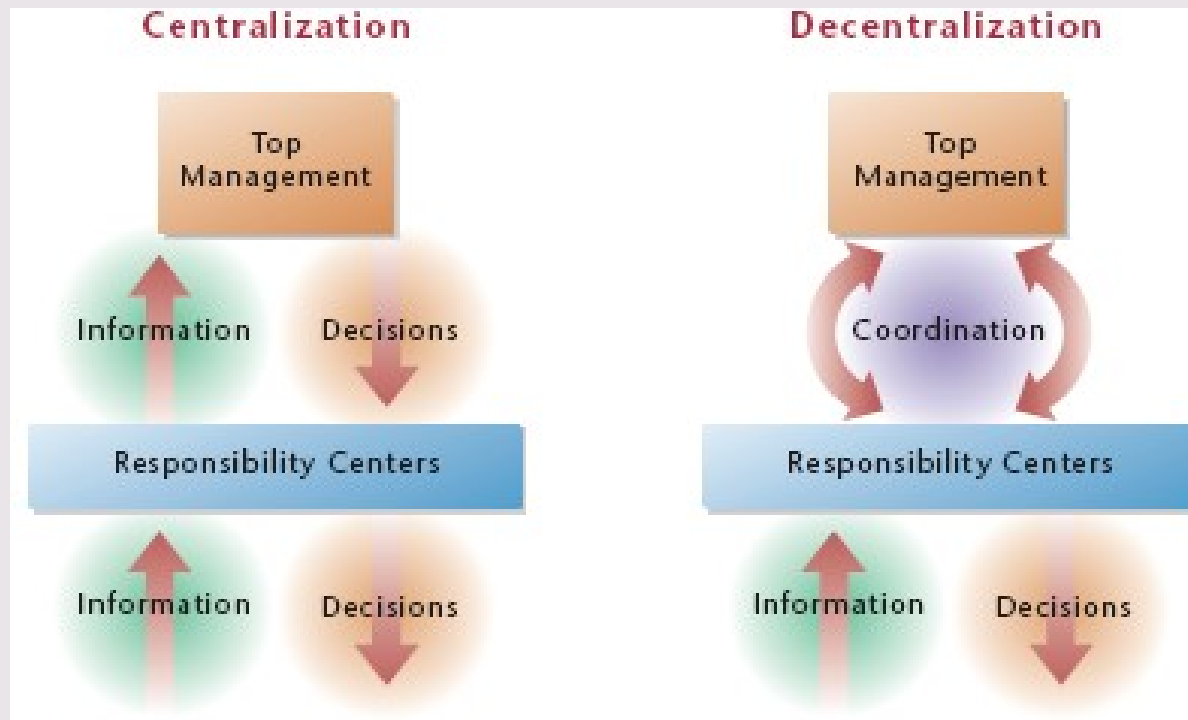


## How do centralized and decentralized firms differ?

In **centralized firms**, decision making occurs at top levels, implementation at lower levels. **Decentralized firms** allow lower-level managers to make and implement decisions.



# CENTRALIZATION & DECENTRALIZATION



**EXHIBIT 10-1**

# REASONS FOR DECENTRALIZATION

Firms decide to decentralize:

- ✳ For ease of gathering, using local information
- ✳ To focus central management
- ✳ To train & motivate segment managers,
- ✳ To enhance competition & expose segments to market forces



# DIVISIONS IN DECENTRALIZED FIRM

Decentralization achieved by creating divisions by

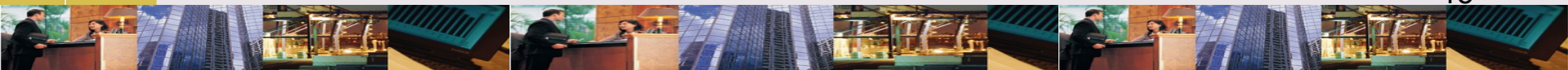
- \* Type of goods & services
- \* Geographic lines
- \* Type of responsibility given to divisional manager



# RESPONSIBILITY CENTER:

## Definition

Is a segment of the business whose manager is accountable for specified sets of activities.



# RESPONSIBILITY CENTERS

Major types of responsibility centers are:

- \* Cost centers

  - \* Manager responsible for cost only

- \* Revenue center

  - \* Manager responsible for sales only

- \* Profit center

  - \* Manager responsible for sales & costs

- \* Investment center

  - \* Manager responsible for sales, costs, & capital investment

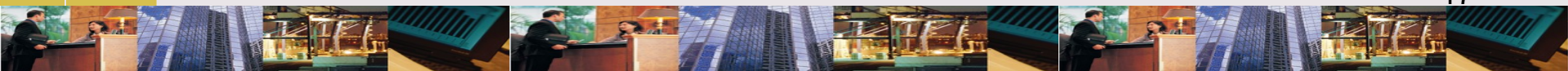




# ***LEARNING OBJECTIVE***

2

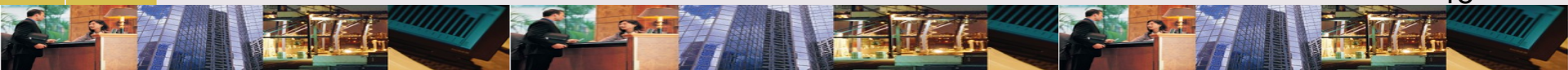
Explain the difference between absorption & variable costing, & prepare segmented income statements.



What are 2 ways to calculate income & how do they differ?

**2** ways to calculate income are by **absorption** costing & **variable** costing.

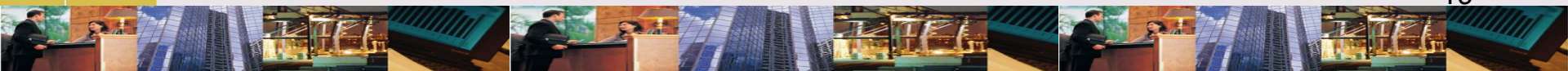
They differ in the treatment of fixed factory overhead.



# INVENTORY VALUATION:

## Background

|                                  |         |
|----------------------------------|---------|
| Units in beginning inventory     | 0       |
| Units produced                   | 10,000  |
| Units sold (\$300 per unit)      | 8,000   |
| <u>Variable costs per unit</u>   |         |
| Direct materials                 | \$ 50   |
| Direct labor                     | 100     |
| Variable overhead                | 50      |
| <u>Fixed costs</u>               |         |
| Fixed overhead per unit produced | 25      |
| Fixed selling & administrative   | 100,000 |



# ABSORPTION COSTING

|                                  |               |
|----------------------------------|---------------|
| Direct materials                 | \$ 50         |
| Direct labor                     | 100           |
| Variable overhead                | 50            |
| Fixed overhead per unit produced | <u>25</u>     |
| Unit product cost                | <u>\$ 225</u> |

Value of ending inventory =  
 $2,000 \times \$ 225 = \text{\textcolor{red}{\$ 450,000}}$

# VARIABLE COSTING

|                   |               |
|-------------------|---------------|
| Direct materials  | \$ 50         |
| Direct labor      | 100           |
| Variable overhead | <u>50</u>     |
| Unit product cost | <u>\$ 200</u> |

Value of ending inventory =  
 $2,000 \times \$ 200 = \$ 400,000$

# COMPARATIVE INCOME STATEMENTS

Income lower under variable costing where fixed costs are expensed for period.

| Fairchild Company<br>Absorption-Costing Income Statement                                                                                                                        |                    |                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------|
| Sales ( $\$300 \times 8,000$ )                                                                                                                                                  |                    | \$2,400,000       |
| Less: Cost of goods sold*                                                                                                                                                       |                    | <u>1,800,000</u>  |
| Gross margin                                                                                                                                                                    |                    | \$ 600,000        |
| Less: Selling and administrative expenses                                                                                                                                       |                    | <u>100,000</u>    |
| Operating income                                                                                                                                                                |                    | <u>\$ 500,000</u> |
| *Using the absorption unit product cost computed in Exhibit 10-5,<br>Cost of goods sold = Absorption unit product cost $\times$ Units sold = $\$225 \times 8,000 = \$1,800,000$ |                    |                   |
| Fairchild Company<br>Variable-Costing Income Statement                                                                                                                          |                    |                   |
| Sales ( $\$300 \times 8,000$ )                                                                                                                                                  |                    | \$2,400,000       |
| Less variable expenses:                                                                                                                                                         |                    |                   |
| Variable cost of goods sold**                                                                                                                                                   | <u>\$1,600,000</u> |                   |
| Contribution margin                                                                                                                                                             |                    | \$ 800,000        |
| Less fixed expenses:                                                                                                                                                            |                    |                   |
| Fixed overhead                                                                                                                                                                  | \$ 250,000         |                   |
| Fixed selling and administrative                                                                                                                                                | <u>100,000</u>     | <u>350,000</u>    |
| Operating income                                                                                                                                                                |                    | <u>\$ 450,000</u> |
| **Using the variable unit product cost computed in Exhibit 10-5,<br>Cost of goods sold = Variable unit product cost $\times$ Units sold = $\$200 \times 8,000 = \$1,600,000$    |                    |                   |

**Exhibit 10-6** Fairchild Company Income Statements Under Absorption and Variable Costing

**EXHIBIT 10-6**

# ABSORPTION INCOME STATEMENT

|                         |                   |
|-------------------------|-------------------|
| Sales (\$300 x 8,000)   | \$ 2,400,000      |
| Less Cost of goods sold | <u>1,800,000</u>  |
| Gross margin            | \$ 600,000        |
| Less S&A expenses       | <u>100,000</u>    |
| Operating income        | <u>\$ 500,000</u> |

CGS =

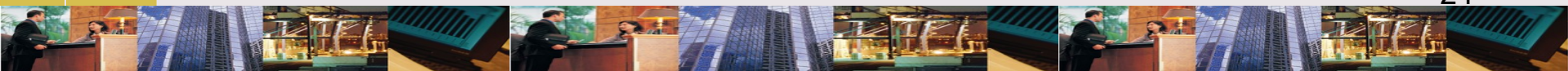
$$8,000 \times \$ 225 = \$ 1,800,000$$

# VARIABLE INCOME STATEMENT

|                        |                   |
|------------------------|-------------------|
| Sales                  | \$ 2,400,000      |
| Less variable expenses | <u>1,600,000</u>  |
| Contribution margin    | 800,000           |
| Less fixed costs       | <u>350,000</u>    |
| Operating income       | <u>\$ 450,000</u> |

Variable costs:  $8,000 \times \$200$

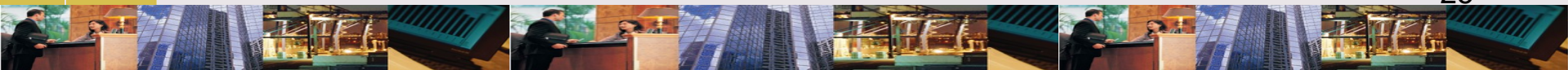
Fixes costs:  $\$250,000 + 100,000$





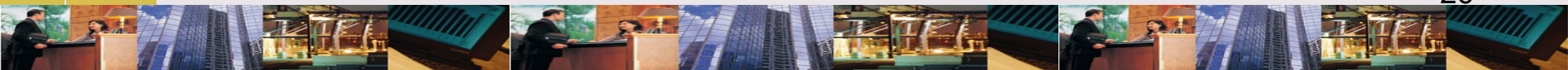
# ABSORPTION VS. VARIABLE

If more is sold than produced, variable costing income  $>$  absorption-costing income, opposite of Fairchild situation. Equal production & sales means equal income.



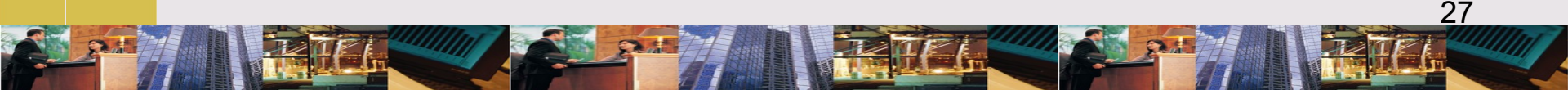
# EXPLANATION

The difference between variable costing & absorption costing year to year is equal to the change in fixed overhead. Under absorption costing, fixed overhead is assigned to inventory produced. Under variable costing, fixed overhead is a period expense.



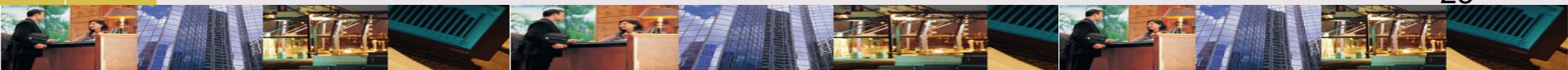
How do variable & absorption costing affect performance evaluation?

**Variable costing** ensures that direct relationship between sales & income holds whereas **absorption costing** does not.



# SEGMENT: Definition

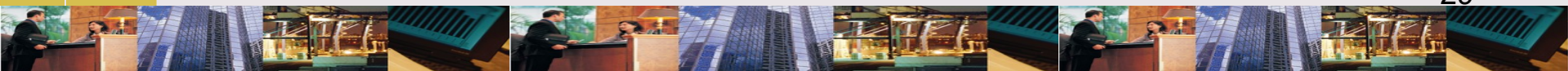
Is a subunit of a company of sufficient importance to warrant performance reports.



# DIRECT FIXED EXPENSES:

## Definition

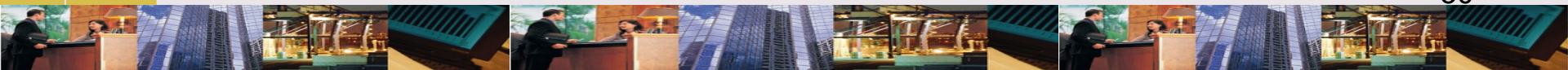
Are fixed expenses directly traceable to a segment & therefore, avoidable. If segment eliminated, so are expenses.



# COMMON FIXED EXPENSES:

## Definition

Are jointly caused by 2 or more segments. These expenses persist even if 1 segment is eliminated.



# COMPARATIVE INCOME STATEMENTS

## Audiomatronics, Inc. Segmented Income Statement For the Coming Year

|                                   | MP3 Players     | DVD Players     | Total             |
|-----------------------------------|-----------------|-----------------|-------------------|
| Sales                             | \$ 400,000      | \$ 290,000      | \$ 690,000        |
| Variable cost of goods sold       | (200,000)       | (150,000)       | (350,000)         |
| Variable selling expense          | <u>(20,000)</u> | <u>(14,500)</u> | <u>(34,500)</u>   |
| Contribution margin               | \$ 180,000      | \$ 125,500      | \$ 305,500        |
| Less direct fixed expenses:       |                 |                 |                   |
| Direct fixed overhead             | (30,000)        | (20,000)        | (50,000)          |
| Direct selling and administrative | <u>(10,000)</u> | <u>(15,000)</u> | <u>(25,000)</u>   |
| Segment margin                    | \$ 140,000      | \$ 90,500       | \$ 230,500        |
| Less common fixed expenses:       |                 |                 |                   |
| Common fixed overhead             |                 |                 | (100,000)         |
| Common selling and administrative |                 |                 | <u>(20,000)</u>   |
| Net income                        |                 |                 | <u>\$ 110,500</u> |

Segment margin is contribution to firm's common fixed costs.

**Exhibit 10-11** Segmented Income Statement

**EXHIBIT 10-11**

# ***LEARNING OBJECTIVE***

3

Compute & explain  
return on investment  
(ROI).

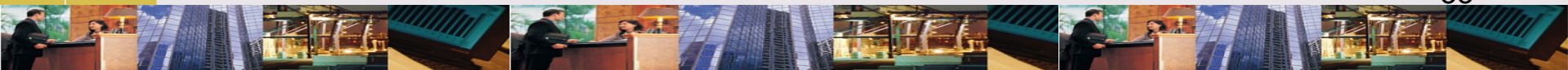


# FORMULA: ROI

*ROI relates operating profits to assets employed.*

**Return on Investment (ROI)**

$$= \frac{\text{Operating Income}}{\text{Average Operating Assets}}$$

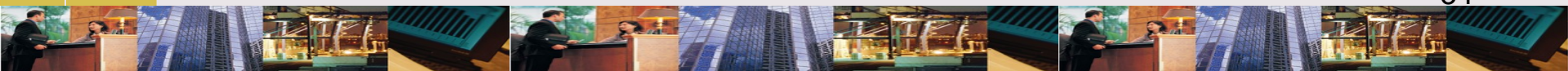


What is operating income?

What are operating assets?

**Operating income** is earnings before interest & taxes.

**Operating assets** are assets acquired to generate operate income.



# ALPHA CO. & BETA CO.

## Background

|                  | Alpha      | Beta        |
|------------------|------------|-------------|
| Operating income | \$ 100,000 | \$ 200,000  |
| Operating assets | \$ 500,000 | \$2,000,000 |

# COMPARING ROI

## ROI: ALPHA

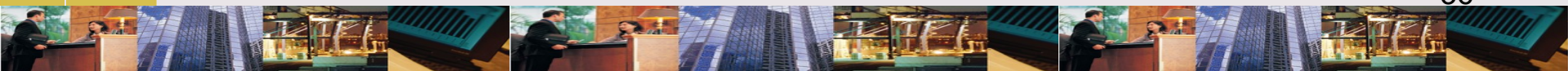
= **Op. Income / Ave. Op. Assets**

= \$100,000 / \$500,000 = **.20**

## ROI: BETA

= **Op. Income / Ave. Op. Assets**

= \$200,000 / \$2,000,000 = **.10**

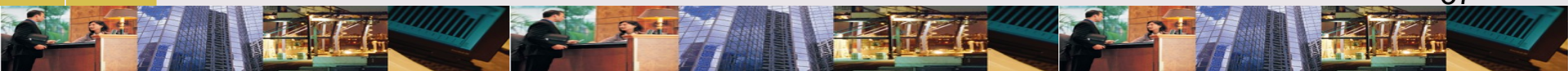


# MARGIN & TURNOVER: ROI

*Separating ROI into margin & turnover provides better analysis.*

**Return on Investment (ROI)**

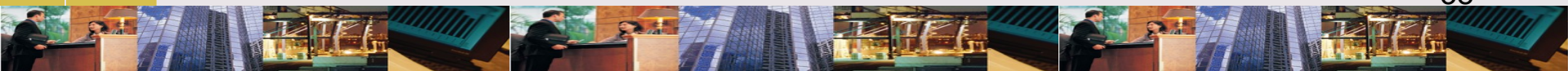
**= (Op. Income / Sales) x (Sales / Ave. Op. Assets)**



What is margin?  
What is turnover?

**Margin** is the ratio of operating to sales.

**Turnover** tells how many dollars of sales results from every dollar of invested assets.



# CELIMAR CO. Background

|                  |            |
|------------------|------------|
| Sales            | \$ 480,000 |
| Operating income | \$ 48,000  |
| Operating assets | \$ 300,000 |

# MARGIN & TURNOVER: ROI

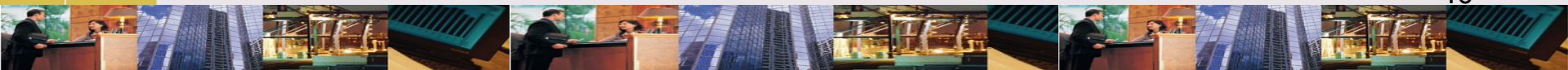
*Separating ROI into margin & turnover  
provides better analysis.*

## Return on Investment (ROI)

$$= (\$48,000 / \$480,000) \times (\$480,000 / \$300,000)$$

$$= 0.10 \times 1.6$$

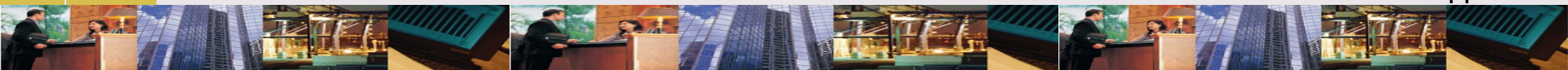
$$= 16\%$$





# EXPLANATION: ROI

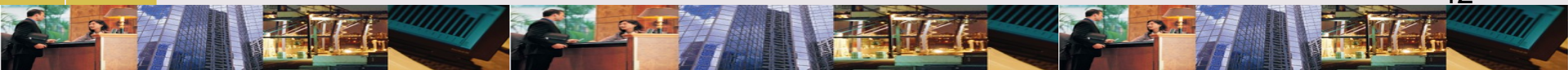
The net return on investments is driven by 2 independent items: the ability to squeeze profit from sales and the ability to squeeze sales from invested assets.



# ADVANTAGES OF ROI

Encourages managers to focus on

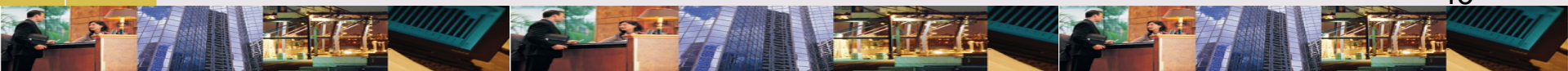
- \* Relationship among sales, expenses (& possibility investment if this is investment center)
- \* Cost efficiency
- \* Operating asset efficiency



# PLASTICS DIVISION EXAMPLE

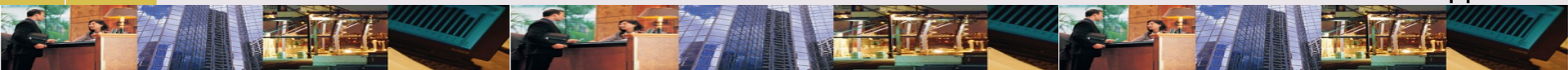
|                  | Without Increased Advertising | With Increased Advertising |
|------------------|-------------------------------|----------------------------|
| Sales            | \$ 2,000,000                  | \$ 2,200,000               |
| Less expenses    | <u>1,850,000</u>              | <u>2,040,000</u>           |
| Operating income | <u>\$ 150,000</u>             | <u>\$ 160,000</u>          |
| Operating assets | \$ 1,000,000                  | \$ 1,050,000               |
| <b>ROI</b>       | <b>15%</b>                    | <b>15.24%</b>              |

The current ROI is the hurdle rate used to make decisions about changes.



# DISADVANTAGES OF ROI

- ✳ Can produce a narrow focus on divisional profitability at expense of profitability for overall firm
- ✳ Encourages managers to focus on short run at expense of long run



# ALTERNATIVES: ROI

|            | <b>Only<br/>Project I</b> | <b>Only<br/>Project II</b> | <b>Both<br/>Projects</b> | <b>Neither<br/>Project</b> |
|------------|---------------------------|----------------------------|--------------------------|----------------------------|
| Op. income | \$ 8,800,000              | \$ 8,140,000               | \$9,440,000              | \$ 7,500,000               |
| Op. assets | \$60,000,000              | \$54,000,000               | \$64,000,000             | \$50,000,000               |
| ROI        | 14.67%                    | <b>15.07%</b>              | 14.75%                   | 15.00%                     |

# ***LEARNING OBJECTIVE***

**4**

Compute & explain  
residual income &  
economic value added  
(EVA).

# RESIDUAL INCOME

*Residual income is the difference between operating income and minimum dollar return on sales.*

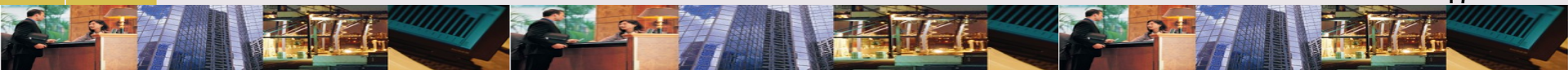
## **Residual Income**

**= Operating income**

**– (Min. rate of return x Ave. Operating Assets)**

**= \$48,000 – (0.12 x \$300,000)**

**= \$12,000**

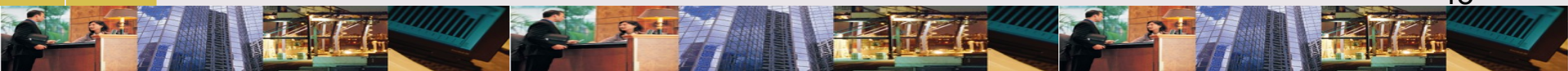


# ALTERNATIVES: Residual Income

In 000s

|               | Only<br>Project I | Only<br>Project II | Both<br>Projects | Neither<br>Project |
|---------------|-------------------|--------------------|------------------|--------------------|
| Op. income    | \$ 8,800          | \$ 8,140           | \$9,440          | \$ 7,500           |
| Op. assets    | \$60,000          | \$54,000           | \$64,000         | \$50,000           |
| Min. return*  | 6,000             | 5,400              | 6,400            | 5,000              |
| Residual Inc. | \$2,800           | \$ 2,740           | <b>\$ 3,040</b>  | \$ 2,500           |

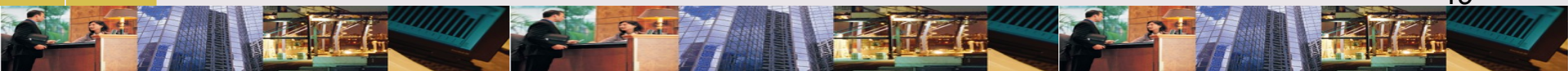
\* 10%





# ADVANTAGES & DISADVANTAGES: Residual Income

- ✱ Advantage: Gives another view of project profitability
- ✱ Disadvantages
  - ✱ Can encourage short run orientation
  - ✱ Direct comparisons are difficult



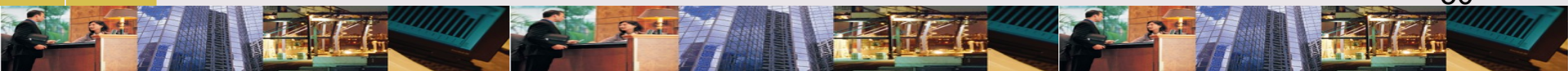
# ECONOMIC VALUE ADDED (EVA)

*EVA is net income minus total annual cost of capital. Projects with positive EVA are acceptable.*

**Economic value added (EVA)**

**= Net income**

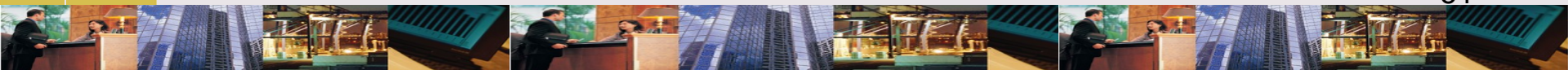
**– (% cost of capital x Capital employed)**



# ***LEARNING OBJECTIVE***

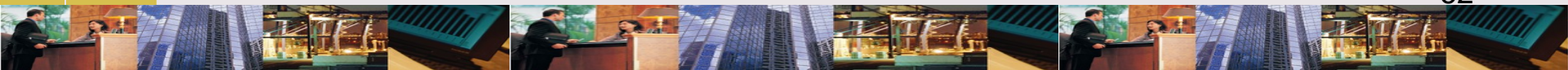
5

Explain the role of transfer pricing in a decentralized firm.



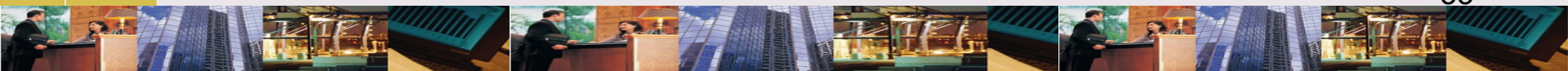
# TRANSFER PRICING: Definition

Is the price charged for a component by the selling division to the buying division of the same company.



What are the minimum & maximum transfer prices?

The **minimum** transfer price would leave the selling division not worse off and the **maximum** would leave the buying division no worse off than if sold (acquire) externally.



# TRANSFER PRICE: Choices

## ✧ Market price

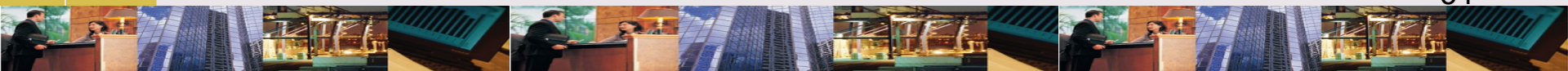
- ✧ Best choice if there is a competitive outside market

## ✧ Cost-Based price

- ✧ When there is not good outside price

## ✧ Negotiated price

- ✧ Useful with there are market imperfections



# CHAPTER 10

***THE END***